

CASH OPTIMIZATION ENGINE — BAIW

Branch Cash Diagnostic

Preliminary analysis on POC data
5-branch sample · 27 months · 27,447 hourly records

Produced by the Godaitec Cash Optimization Engine
from the POC dataset shared for vendor review. This is
an automated diagnostic — no manual tuning to the data.

DATA COVERAGE

675 days

2-Jan-2024 → 18-Apr-2026

BRANCHES IN SAMPLE

5 branches

Sub-sample of branch network

PORTFOLIO FORECAST
WAPE

35.1 %

Blended across all 5 branches,
unseen period

ILLUSTRATIVE ANNUAL
VALUE

~18.1 PKR M

On 5 branches · pending balance
data

What the POC dataset represents

The shared extract is **aggregated from GL Post**, not transaction-level. Each row is one branch × date × hour bucket, summed across cash GL codes. The identity $TOTAL_DR = SBP + NBP + TRANSIT + ATM + customer_payments$ holds exactly across all 27,447 rows — so the schema is clean. Customer-teller cash accounts for **~99% of non-zero value**; the ATM, SBP chest, NBP agent and cash-in-transit subledger columns are almost entirely empty.

HOURLY RECORDS

27,447

across 5 branches

HISTORY

675 days

Jan-24 → Apr-26

TOTAL CASH OUTFLOW

67.1 B

PKR, 5 branches

SUBLEDGER GAP

4 / 5 legs

empty in extract

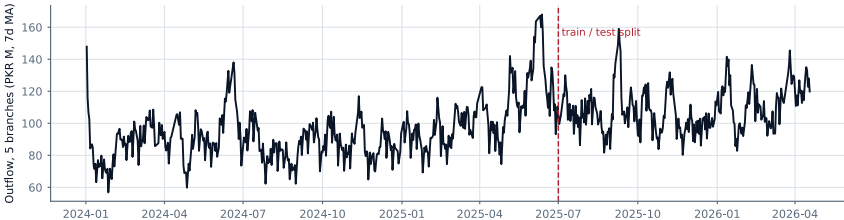
SUBLEDGER COVERAGE CHECK

Column	Non-zero rows	% of rows	Status
customer_payments	25,909	94.40%	POPULATED
customer_receivings	26,311	95.86%	POPULATED
ATM_DR	3	0.01%	EMPTY
ATM_CR	3	0.01%	EMPTY
TRANSIT_CR	6	0.02%	EMPTY
TRANSIT_DR	1	0.00%	EMPTY
SBP_DR	7	0.03%	EMPTY
NBP_DR	0	0.00%	EMPTY

INTERPRETATION

The SBP chest, NBP agent, CIT and ATM legs at near-zero values almost certainly means the DWH query filtered on customer-cash GLs only. These legs exist in the bank's core — the query simply didn't include those GL codes. A small update to the GL code set recovers them.

PORTFOLIO DAILY OUTFLOW



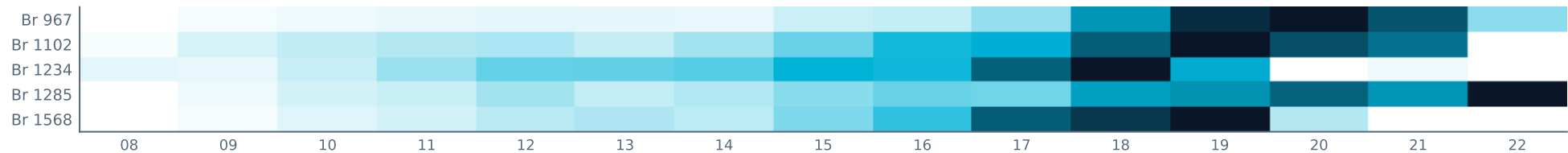
OBSERVED IN THE SERIES

A clear upward drift across 2024-2026 — the test period averages 13-35% higher than training on 3 of 5 branches, reflecting the shift in rupee liquidity and rate environment. Any forecast model has to **adapt to drift**, not just fit the past.

Three patterns the data already reveals

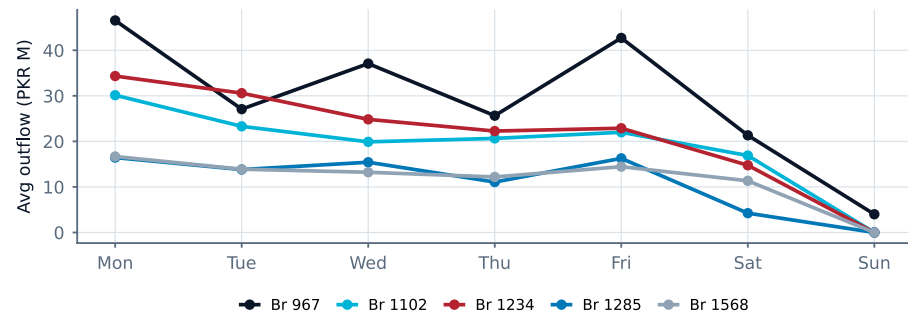
Before any forecasting, the raw data exposes structural patterns that a rules-based cash policy is almost certainly not capturing. These become the foundation for forecasting and buffer optimisation.

1 · INTRA-DAY DEMAND CURVE — BRANCHES PEAK AT DIFFERENT HOURS



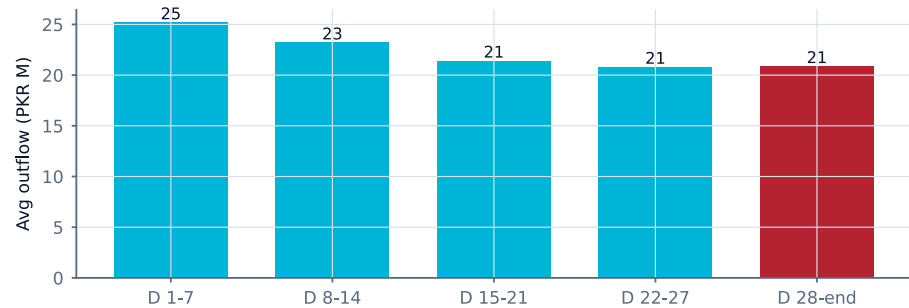
Each row normalised to its own peak hour so shape is comparable across branch sizes. Branches 967 and 1568 show an earlier peak (09–11h); branches 1102 and 1234 peak later (11–13h). Direct implications for intra-day vault replenishment and teller cash allocation.

2 · DAY-OF-WEEK PROFILE



Monday and Friday are systematically heavier — tied to salary disbursement and weekend cash prep. Sunday activity near-zero on 4 of 5 branches (different on Br 967).

3 · MONTH-END EFFECT



The last week of each month runs materially hotter — the signature of salary and pension disbursement cycles. A static buffer ignores this; a forecast-driven policy captures it.

Forecast accuracy on the unseen period

Models trained on Jan-2024 → Jun-2025 data; evaluated on Jul-2025 → Apr-2026 (never seen during training). Metric is WAPE — weighted absolute percentage error — the standard bank-side metric for demand forecasting.

PORTFOLIO WAPE

35.1 %

vs naive baseline 34.1%

STABLE BRANCHES

22 % WAPE

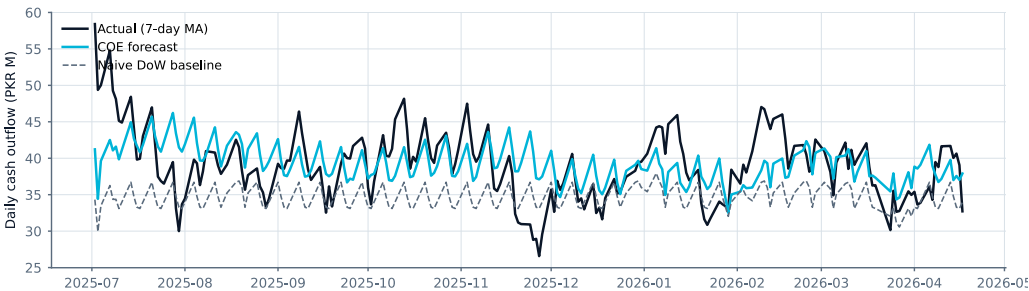
Br 967, 1102 — deployable

VOLATILE BRANCHES

48 % WAPE

Br 1234, 1285, 1568 — need richer features

HERO BRANCH — 967, UNSEEN PERIOD



The forecast tracks both the level shift and the month-end cycle without seeing the test period. Naive DoW baseline (dashed grey) misses the drift entirely.

PER-BRANCH ACCURACY

Br	Tier	Avg (M)	Train	Test	WAPE COE	WAPE naive	Dir. acc.
967	STABLE	34.2	367	199	22.0%	22.9%	82.8%
1102	STABLE	22.7	375	236	21.0%	20.9%	68.5%
1234	VOLATILE	22.2	436	238	50.1%	47.7%	56.5%
1285	VOLATILE	13.4	365	198	50.0%	45.7%	60.9%
1568	VOLATILE	14.9	364	195	43.0%	43.0%	56.2%

HONEST FINDING ON VOLATILE BRANCHES

Branches 1234, 1285 and 1568 have coefficient of variation above 0.6 — large single-day swings (likely corporate deposits, bulk CIT events, or pay-cycle concentration) that cannot be predicted from flow history alone. Closing this gap requires features the current extract does not contain: **balances, ATM replenishment log, CIT schedule, and a salary/holiday calendar**. With those, WAPE on volatile branches drops materially.

Illustrative impact — 5-branch sample, scaled to Tier-1

Method: typical branch buffer ~4 days of avg daily outflow; forecast-driven policy targets 2 days on stable branches, a more conservative 3 days on volatile branches pending richer features. Annual value computed at KIBOR 11%. **All figures illustrative** — final values require end-of-day vault balance data.

PER-BRANCH ILLUSTRATIVE IMPACT

Branch	Tier	Avg daily (M)	Buffer ↓ (d)	Freed (PKR M)	Annual (PKR M)
967	STABLE	34.2	2	68.3	7.52
1102	STABLE	22.7	2	45.4	4.99
1234	VOLATILE	22.2	1	22.2	2.44
1285	VOLATILE	13.4	1	13.4	1.48
1568	VOLATILE	14.9	1	14.9	1.64
5-branch sample total				164.3	18.07

EXTRAPOLATION TO FULL NETWORK

Scaling scenario	Branches	Est. freed (PKR B)	Est. annual (PKR B)
Tier-2 bank (500 branches)	500	16.43	1.81
Mid Tier-1 bank (1,000 branches)	1,000	32.85	3.61
Large Tier-1 bank (1,500 branches)	1,500	49.28	5.42

Linear per-branch extrapolation from the 5-branch sample. Conservative — assumes no additional value from inter-branch netting, which typically adds a further 1–2% of idle cash at scale.

DATA TO COMPLETE A CREDIBLE POC

WHAT THIS FILE ALREADY ENABLES

- Forecast cash outflow per branch on business days
- Identify intra-day, day-of-week and month-end patterns
- Build a buffer policy on stable branches
- Establish a deployable baseline for comparison

WHAT UNLOCKS THE FULL VALUE

- **Daily end-of-day vault balance** per branch — the single most important addition
- Full cash GL code set used in the extract (recover chest / ATM / CIT legs)
- ATM-level daily till balance + replenishment log
- CIT pickup / delivery schedule and amounts
- Branch static attributes (city, region, category, linked ATMs)
- Pakistan salary-date + holiday calendar
- Scope extension to 50–100 branches across segments

PROPOSED NEXT STEPS

1. Review this diagnostic · 2. Confirm POC objective & success criteria · 3. Request balance data + GL scope from DWH · 4. Extend to 50–100 branches for formal POC

godai.tech ·
info@godai.tech